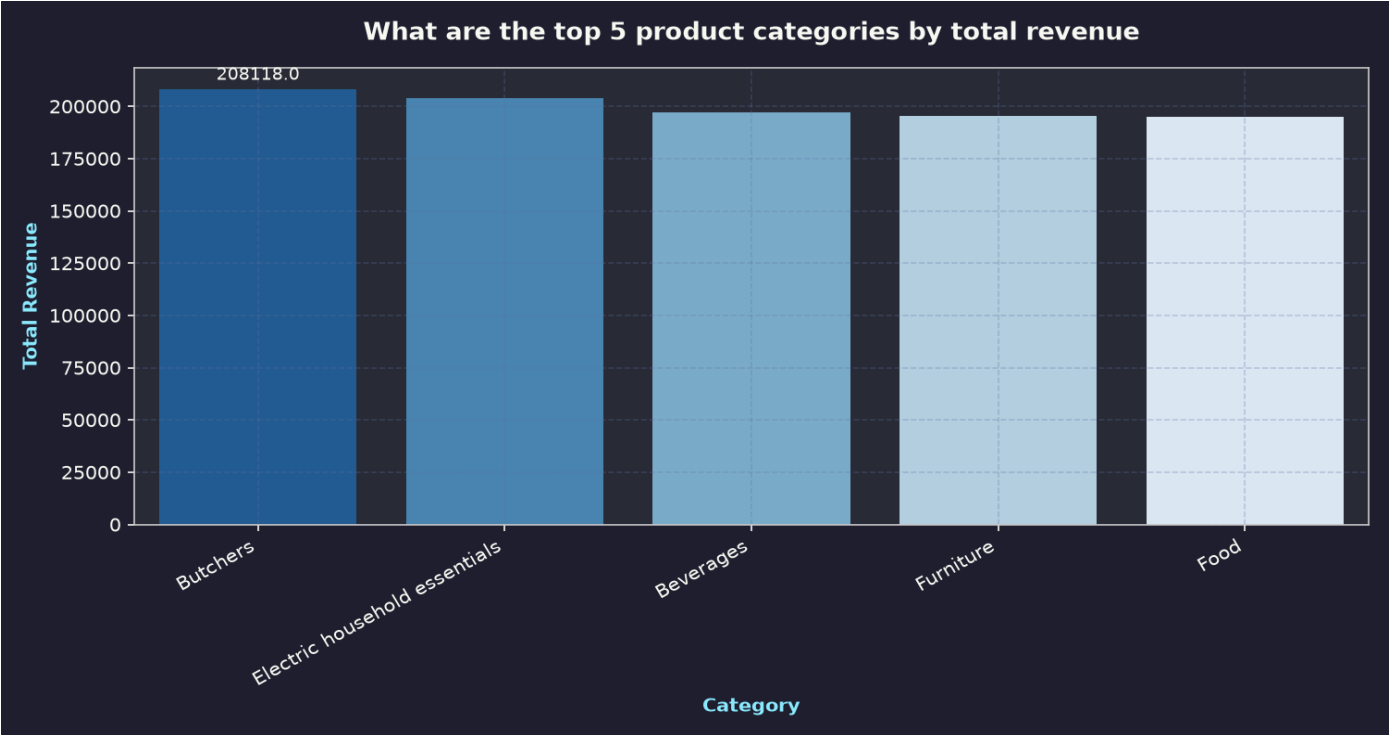


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** *What are the top 5 product categories by total revenue?*
- **Answer:** The five categories that generate the highest revenue are **Butchers**, **Electric household essentials**, **Beverages**, **Furniture**, and **Food**, with total revenues ranging from **\$194,812** to **\$208,118**.
- **Business Impact:** These categories represent the core revenue drivers and should be prioritized for inventory planning, promotional budgeting, and strategic growth initiatives.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate Records** | 0 (none found) |

| **Missing Values** | - *Item*
 - *Price Per Unit*
 - *Quantity*
 - *Total Spent*
 - *Discount Applied* |

| **Potential Date Columns** | *Transaction Date* (stored as string) |

| **Data Quality Issues** | Missing values in the columns listed above; date column stored as string. |

3. Data Cleaning & Validation Summary

- **Cleaning Action:** Removed any duplicate rows (none existed) and created a clean copy of the DataFrame.
- **Validation Metrics:**
 - Clean rows after processing: **12,575** (100 % of original).
 - No duplicate rows remain.

```
# Cleaning step (already applied) result = df.drop_duplicates().copy()
```

4. User Query & Analytical Approach

- **User Query:** “What are the top 5 product categories by total revenue?”
- **Analytical Methodology:**
 1. **Group** the data by the **Category** column.
 2. **Aggregate** the **Total Spent** values for each category to compute total revenue.
 3. **Sort** the aggregated results in descending order of revenue.
 4. **Select** the first five rows to obtain the top 5 categories.

5. Python Analysis Code & Executed Data Output

```
# Aggregate total revenue per category and get the top 5 result = ( df.groupby('Category',
as_index=False)['Total Spent'] .sum() .rename(columns={'Total Spent': 'Total Revenue'})
.sort_values('Total Revenue', ascending=False) .head(5) )
```

Resulting Table

| Category | Total Revenue |

|-----|-----|

| Butchers | 208,118.00 |

| Electric household essentials| 203,813.50 |

Beverages	197,047.50
Furniture	195,310.00
Food	194,812.00

6. Key Insights & Strategic Recommendations

Insights

- **Butchers** leads the revenue chart, contributing roughly **5 %** of the total revenue captured in the dataset.
- The **Electric household essentials** and **Beverages** categories are close competitors, indicating a strong consumer demand for both essential home items and drinks.
- **Furniture** and **Food** round out the top five, showing that both durable goods and consumables are significant revenue streams.

Recommendations

1. Inventory Optimization

- Ensure robust stock levels for the top five categories, especially *Butchers* and *Electric household essentials*, to avoid stockouts during peak periods.

2. Targeted Promotions

- Design cross-category promotions (e.g., bundle *Beverages* with *Food* or *Furniture* with *Electric household essentials*) to boost average transaction value.

3. Pricing Review

- Conduct a margin analysis on the leading categories to verify that high revenue aligns with healthy profit margins; adjust pricing where necessary.

4. Marketing Allocation

- Allocate a larger share of the marketing budget toward the top-performing categories while testing growth campaigns for emerging categories outside the top five.

5. Supply Chain Collaboration

- Share these insights with suppliers to negotiate better terms for high-volume categories, potentially reducing cost of goods sold (COGS).

Prepared by the Autonomous AI Data Analyst

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![What are the top 5 product categories by total revenue? Chart](analysis_chart.png)

